

Programming Reserved Register Field

Introduction

A field in a register is reserved if the field is not open to users, or produces unpredictable results if configured to values other than defaults.

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The reserved fields should not be modified. It is not possible to write only part of a register since registers must always be written as a whole. As a result, to write an entire register that contains reserved fields, you can choose one of the following two options:

1. Read the value of the register, modify only the fields you want to configure and then write back the value so that reserved fields are untouched.

OR

2. Modify only the fields you want to configure and write back the default value of the reserved fields. The default value of a field is provided in the "Reset" line of a register diagram. For example, the default value of Field_A in [Register X](#) is 1.

Register 63.1. Register X (Address)

(reserved)												Field_C				(reserved)												Field_B		Field_A					
31												20		19		16		15														2		1	0
0 0 0 0 0 0 0 0 0 0 0 0												0000				0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0												0	1	Reset					

Suppose you want to set Field_A, Field_B, and Field_C of [Register X](#) to 0x0, 0x1, and 0x2, you can:

- Use option 1 and fill in the reserved fields with the value you have just read. Suppose the register reads as 0x0000_0003. Then, you can modify the fields you want to configure, thus writing 0x0002_0002 to the register.
- Use option 2 and fill in the reserved fields with their defaults, thus writing 0x0002_0002 to the register.